

magnitude as shown above.

22 D total energy delivered by the battery = $2500 + 500 = 3000 \text{ J}$

$$\text{e.m.f.} = \frac{\text{total energy}}{\text{total charge}} = \frac{3000}{1500} = 2.00 \text{ V}$$

23 C Increasing R_1 increases the effective resistance of the combination of R_0 and R_1 in parallel